Track 1: Data Dashboarding

Key Deliverables for Track 1:

- Dashboard File (.pbix, .twbx, .xlsx)
- PDF Export or high-resolution dashboard screenshots
- One-page Business Insights Summary detailing actionable recommendations

Problem Statement [D1]
Supply Chain Carbon Footprint & ESG Tracker

Brief & Case Scenario:
Global logistics companies are under immense pressure to track and reduce their Scope 3 emissions. Currently, data regarding transport modes (Air, Sea, Road), distance traveled, and fuel consumption is siloed, making it impossible to see the "true carbon cost" of a shipment. The challenge is to build a dashboard that calculates Total CO2e per shipment and identifies the most polluting routes or vendors.

Outcome:
A comprehensive ESG dashboard that provides complete visibility into shipping emissions, enabling logistics managers to strategically re-route shipments and confidently meet corporate carbon reduction targets.

Dataset:
A synthetic dataset of 5,000 shipments, including origin/destination, transport mode, weight, and standard emission conversion factors.

Tools:
Power BI, IBM Cognos Analytics, Looker Studio or Excel.

Constraints:
Must include a dynamic calculation to convert distance and weight into a unified CO2e metric.

Key Features:

- **Emission Heatmap:**
  Visual representation of high-emission global shipping routes.
- **Vendor Scorecard:** Ranking 3PL (third-party logistics) providers based on their carbon efficiency.
- **What-If Scenario Simulator:** A tool to calculate emission savings if 20% of Air freight is shifted to Sea freight.
- **YoY Reduction Tracker:** A progress bar against a corporate goal of a 15% emission reduction.

Problem Statement [D2]
SaaS Cloud Infrastructure Cost Optimizer (FinOps)

Brief & Case Scenario:
Fast-growing tech startups often over-provision their cloud infrastructure (AWS/GCP/Azure), leading to massive "cloud waste." Engineering managers need a visual tool to identify "zombie" instances (servers running but doing nothing) and underutilized databases so they can confidently scale down and save money.

Outcome:
A targeted FinOps dashboard that highlights wasted cloud spend, empowering engineering teams to right-size their infrastructure and significantly reduce their monthly burn rate without sacrificing performance.

Dataset:
Synthetic cloud billing export containing instance IDs, CPU/Memory utilization percentages, hourly cost, uptime, and region.

Tools:
Power BI, Tableau, IBM Cognos Analytics, Looker Studio or Excel.

Constraints:
Focus specifically on identifying resources operating below 10% CPU utilization for more than 72 hours.

Key Features:

- **Idle Resource Flagger:** A prioritized table of the most expensive "zombie" instances to shut down immediately.
- **Cost by Microservice:** Grouping raw server costs into human-readable project or department buckets.
- **Projected Burn Rate:** A forecast of the month-end cloud bill based on the current daily run rate.
- **Instance Right-Sizing Recommendations:** Visual alerts suggesting a downgrade from 'Large' to 'Medium' instances based on historical memory usage.

Problem Statement [D3]
Healthcare Telemedicine Triage & Bottleneck Monitor

Brief & Case Scenario:
A major telehealth platform experiences severe patient drop-offs during peak flu season because patients spend too long in the "virtual waiting room." The hospital administration needs to visualize the patient journey from login to doctor consultation to find out exactly where the bottleneck occurs and how to dynamically reallocate on-call doctors.

Outcome:
A clinical operations dashboard that tracks real-time patient flow, allowing hospital administrators to reduce abandonment rates and optimize doctor utilization across different medical specialties.

Dataset:
Patient session logs including triage severity (Red/Yellow/Green), queue entry timestamp, consultation start/end timestamps, and doctor specialty.

Tools:
Power BI, Tableau, IBM Cognos Analytics, Looker Studio or Excel.

Constraints:
Must calculate and visualize the "Time-in-Queue" delta using the provided timestamps.

Key Features:

- **Real-time Queue Status:** A visual indicator of how many patients are waiting per specialty (e.g., Pediatrics vs. General Practice).
- **Patient Abandonment Curve:** A line chart correlating the length of wait time with the percentage of patients who log off before seeing a doctor.
- **Peak Hour Heatmap:** Identifying the specific hours and days where the system is most overwhelmed.
- **Staffing Reallocation Prompt:** Highlighting when a specific department's wait time breaches a 15-minute SLA.

Problem Statement [D4]
Retail E-Commerce Dynamic Margin & Cannibalization Analyzer

Brief & Case Scenario: An online retailer is running aggressive flash sales, which are driving up top-line revenue but crushing net profit margins. They suspect certain heavy discounts are "cannibalizing" the sales of full-priced items. The challenge is to build a dashboard that shows the true profitability of promotional campaigns.

Outcome:
A merchandising dashboard that reveals the net profit impact of discounts, helping the pricing team move away from unprofitable flash sales toward highly targeted, data-driven promotions.

Dataset:
Transactional data including base cost, discounted sale price, promo code used, product category, and return rates.

Tools:
Power BI, Tableau, IBM Cognos Analytics, Looker Studio or Excel.

Constraints:
Must distinguish between gross revenue and net profit after the discount and cost-of-goods-sold (COGS) are applied.

Key Features:

- **Profitability Matrix:** A scatter plot mapping sales volume against net margin per product.
- **Discount Effectiveness Gauge:** Showing whether a 30% discount actually resulted in enough volume to offset the lost margin.
- **Return Rate Correlation:** Tracking if highly discounted items have a higher rate of return/refunds.
- **Category Performance Breakdown:** Visualizing which product lines can survive heavy discounting and which cannot.

Problem Statement [D5]
FinTech Transaction Anomaly & Fraud Visualizer

Brief & Case Scenario:
A digital challenger bank is facing an uptick in account takeovers. Traditional tabular reports are too slow for fraud analysts to spot coordinated attacks. They need a visual dashboard that highlights unusual transaction velocities (e.g., 10 small transactions in 5 minutes) or geographic anomalies (e.g., a card used in New York and London within an hour).

Outcome:
A visual risk dashboard that instantly flags suspicious transaction patterns, significantly reducing the time it takes analysts to freeze compromised accounts and preventing financial loss.

Dataset:
Transaction logs containing user IDs, timestamps, merchant category codes, transaction amounts, geolocation data, and status (approved/flagged).

Tools:
Power BI, Tableau, IBM Cognos Analytics, Looker Studio or Excel.

Constraints: Focus on visual anomaly detection rather than building complex predictive ML models (due to the 6-hour limit).

Key Features:

- **Geographic Anomaly Map:** Plotting transactions to visually highlight impossible travel times between card swipes.
- **Transaction Velocity Spikes:** A timeline highlighting user accounts with an abnormal frequency of transactions within a 1-hour window.
- **High-Risk Merchant Categorization:** A pie chart or tree map of fraud incidents grouped by merchant type (e.g., Crypto exchanges, Electronics).
- **False Positive Tracker:** Monitoring how many legitimate transactions were incorrectly flagged by the current rule set.